

17. Suppose the following values are inserted into a binary tree, in the order given:

12, 7, 9, 10, 22, 24, 30, 18, 3, 14, 20

Draw a diagram of the resulting binary tree.

18. How would the values in the tree you sketched for Question 17 be displayed in an inorder traversal?
19. How would the values in the tree you sketched for Question 17 be displayed in a pre-order traversal?
20. How would the values in the tree you sketched for Question 17 be displayed in a postorder traversal?

True or False

21. T F Each node in a binary tree must have at least two children.
22. T F When a node is inserted into a tree, it must be inserted as a leaf node.
23. T F Values stored in the current node's left subtree are less than the value stored in the current node.
24. T F The shape of a binary tree is determined by the order in which values are inserted.
25. T F In inorder traversal, the node's data is processed first, then the left and right nodes are visited.

Programming Challenges

1. Binary Tree Template

Write your own version of a class template that will create a binary tree that can hold values of any data type. Demonstrate the class with a driver program.

2. Node Counter

Write a member function, for either the template you designed in Programming Challenge 1 or the `IntBinaryTree` class, that counts and returns the number of nodes in the tree. Demonstrate the function in a driver program.

3. Leaf Counter

Write a member function, for either the template you designed in Programming Challenge 1 or the `IntBinaryTree` class, that counts and returns the number of leaf nodes in the tree. Demonstrate the function in a driver program.

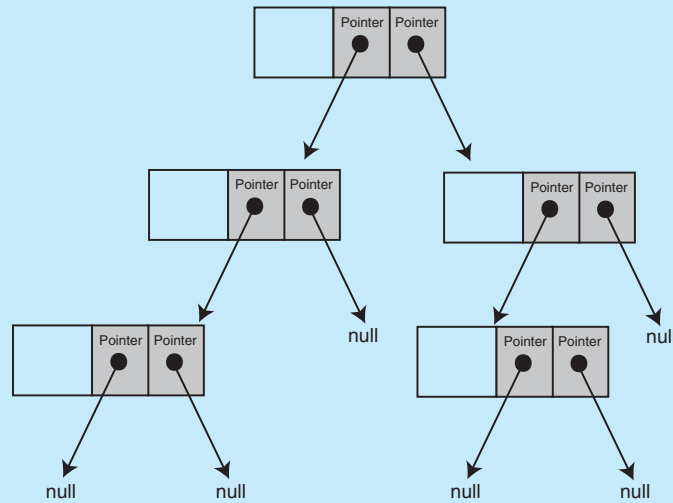
4. Tree Height

Write a member function, for either the template you designed in Programming Challenge 1 or the `IntBinaryTree` class, that returns the height of the tree. The height of the tree is the number of levels it contains. For example, the tree shown in Figure 20-10 has three levels.



VideoNote
Solving the
Node Counter
Problem

Figure 20-10



Demonstrate the function in a driver program.

5. Tree Width

Write a member function, for either the template you designed in Programming Challenge 1 or the `IntBinaryTree` class, that returns the width of the tree. The width of the tree is the largest number of nodes in the same level. Demonstrate the function in a driver program.

6. Tree Assignment Operator and Copy Constructor

Design an overloaded assignment operator and a copy constructor for either the template you designed in Programming Challenge 1 or the `IntBinaryTree` class. Demonstrate them in a driver program.

7. Queue Converter

Write a program that stores a series of numbers in a binary tree. Then have the program insert the values into a queue in ascending order. Dequeue the values and display them on the screen to confirm that they were stored in the proper order.

8. Employee Tree

Design an `EmployeeInfo` class that holds the following employee information:

Employee ID Number: an integer
Employee Name: a string

Next, use the template you designed in Programming Challenge 1 to implement a binary tree whose nodes hold an instance of the `EmployeeInfo` class. The nodes should be sorted on the Employee ID number.

Test the binary tree by inserting nodes with the following information.

Employee ID Number	Name
1021	John Williams
1057	Bill Witherspoon
2487	Jennifer Twain
3769	Sophia Lancaster
1017	Debbie Reece
1275	George McMullen
1899	Ashley Smith
4218	Josh Plemmons

Your program should allow the user to enter an ID number, then search the tree for the number. If the number is found, it should display the employee's name. If the node is not found, it should display a message indicating so.